

Phase 4 - Data Migration, Testing and Security

Project: Leave Management System (ServiceNow Scoped Application)

Scope: `x_1958846_leave_0`

1. Objective

Everything built in Phases 2 and 3 was validated using an admin account with unrestricted access appropriate for building and testing, but not for real, multi-user production use. Phase 4 closed that gap: it hardened the application with genuine role-based security enforced at the platform level (Access Control Lists), replaced the temporary admin-only override logic with proper role checks, established a structured QA pass across the full system, and defined a real path for migrating existing leave data into the new system.

The central theme of this phase was moving every piece of "trust" in the system from *convention* (UI fields being read-only) to *enforcement* (ACLs actually blocking unauthorized access), because UI-level restrictions alone can always be bypassed by a sufficiently motivated user or a direct API call.

2. Roles and Access Control (ACLs)

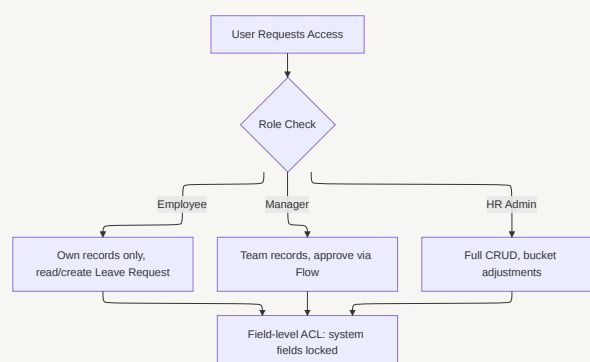
2.1 Roles Implemented

Role	Description
<code>x_1958846_leave_0.employee</code>	Can create and view their own Leave Requests; read-only access to their own Leave Bucket
<code>x_1958846_leave_0.manager</code>	Can view and approve/reject requests where they are the Manager of the Requester; read access to their team's Leave Buckets
<code>x_1958846_leave_0.hr_admin</code>	Full CRUD access on Leave Bucket and Leave Calculator; authorized to create or adjust employee balances
<code>admin</code>	Full system access — retained for platform administration only, no longer used as a stand-in for the manager role in application logic

A key correction made during this phase: the Approve/Reject UI Actions built in Phase 2 originally checked `gs.hasRole('admin')` as a placeholder during initial testing. This was replaced with a proper check against the `x_1958846_leave_0.manager` role, combined with a record-level condition confirming the logged-in user is actually the Manager of the specific request's Requester — closing the gap where any admin, rather than only the correct manager, could act on a request.

2.2 ACL Matrix

Table	Operation	Employee	Manager	HR Admin
Leave Request	Create	✓ (own)	✗	✓
Leave Request	Read	✓ (own)	✓ (team)	✓ (all)
Leave Request	Write	✗ (system fields locked)	✗	✓
Leave Bucket	Read	✓ (own)	✓ (team)	✓ (all)
Leave Bucket	Write	✗	✗	✓
Leave Calculator	Read	✓	✓	✓
Leave Calculator	Write	✗	✗	✓



3. Data Migration

Migrating from spreadsheet-based leave tracking into the new system required careful handling, since the accuracy of every employee's starting balance directly determines whether the system is trusted from day one.

Step	Description
1	Exported existing HR leave balances from the legacy source into a staging CSV structured to match the Leave Bucket table
2	Validated that every Employee reference resolved correctly to an existing <code>sys_user</code> record — using the real underlying <code>user_name</code> field rather than the calculated display Name field, a lesson learned directly during Phase 2 build/test
3	Imported the staged data into Leave Bucket via a proper Import Set and Transform Map , rather than a background script, to get built-in error handling, logging, and a repeatable, auditable migration process
4	Reconciled a representative sample of migrated balances against the original source data to confirm accuracy before go-live
5	Populated the Leave Calculator with real, HR-approved country-wise policy values, replacing the placeholder test data used during Phases 1–3

4. Testing Strategy

4.1 Test Levels

Level	Coverage
Unit	Each Business Rule tested in isolation, confirming it behaves correctly independent of the others
Integration	The full request lifecycle tested end-to-end: submit → route to manager → approve/reject → bucket update
Regression	The full integration suite re-run after every change introduced in Phases 3 and 4, to catch any unintended side effects
Security/ACL	Verified that role restrictions actually block unauthorized actions at the platform level, not merely hide UI elements a determined user could still bypass
UAT	HR Business team validated the completed system directly against the original Phase 1 requirements before sign-off

4.2 Key Test Cases

#	Scenario	Expected Result	Outcome
1	Submit request with no matching Leave Bucket	Blocked with a clear, actionable error message	✓ Pass
2	Submit request exceeding available balance	Blocked, showing requested vs. available days	✓ Pass
3	Half Day request	Duration = 0.5, End Date auto-matches Start Date	✓ Pass
4	End Date before Start Date	Blocked both client-side (instant) and server-side (authoritative)	✓ Pass
5	Approve via Flow-routed manager	Approval field auto-updates; Bucket Taken/Balance updates accordingly	✓ Pass
6	Reject via Flow-routed manager	Approval updates to Rejected; Bucket remains unchanged	✓ Pass
7	Duration exceeding 5 days	Comments field becomes mandatory before submission is allowed	✓ Pass
8	Direct edit attempt on Status/Approval fields	Blocked by both UI Policy and ACL enforcement	✓ Pass
9	Non-manager attempts to approve a request	Blocked by role-based ACL, independent of any UI Action visibility	✓ Pass
10	Duplicate approval prevention	Approve/Reject actions disappear once a decision has already been made	✓ Pass

5. Data Integrity Hardening

Several refinements were made during this phase specifically to close gaps identified through testing:

- Formalized the policy on **Work From Home** as a genuine, balance-deducting Leave Type — treated consistently with all other leave types rather than as a special case, per final HR business decision.

- Added a duplicate-prevention safeguard on Leave Bucket to ensure only one record can exist per Employee + Leave Type combination, preventing balance data from ever being split across duplicate ledgers.
- Documented the known limitation around cancellation of an already-approved request (no automatic balance reversal in v1) as an explicit, tracked backlog item rather than a silent gap — carried forward into the Project Conclusion's future roadmap.
- Confirmed the Leave Calculator's policy values are correctly reflected in the Leave Bucket accrual process used during onboarding of new leave-year balances.